# Randomized and Observational Estimates in a Four-Arm Within-Study Comparison: A Bounded Analysis of Estimand Alignment

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**Abstract**

Within-study comparisons (WSCs) place randomized and nonrandomized estimates in a common empirical setting, but a comparison is informative only when the target parameters and their identifying assumptions are made explicit. Using the publicly released WSCdata mathematics–vocabulary study, this analysis compares two protocol-defined pairs of estimates for mathematics post-test scores. The primary comparison aligns the randomized effect among mathematics-preferring participants with a cross-fitted augmented inverse-probability-weighted (AIPW) average treatment effect on the treated (ATT) in the concordant quasi-experimental-design (QED) subset. The randomized estimate was 0.794682 and the observational ATT estimate was 0.794234; their difference (QED minus RCT) was −0.000448, with a 2,000-replicate joint-bootstrap percentile 95% interval of [−0.949160, 0.663849]. Thus, the point estimates were close in this sample, but the uncertainty does not establish equivalence or precise agreement. A separately specified, owner-decided secondary comparison paired a full-sample randomized ATE with a concordant-subset AIPW ATE. The estimates were 0.760345 and 1.075397, respectively; the difference was 0.315052, with a bootstrap interval of [−0.624904, 1.144430]. The secondary ATE pair cannot be treated as estimating the same parameter because the design probability of one-half was not independently verified. Both bootstrap procedures completed all 2,000 requested replicates, which does not establish interval coverage or causal identification. The evidence supports a bounded within-study comparison, not a general conclusion that observational estimation recovers randomized effects or that estimand alignment explains differences between the two contrasts.

**Keywords:** within-study comparison; randomized controlled trial; quasi-experimental design; estimand alignment; average treatment effect; average treatment effect on the treated; augmented inverse-probability weighting; cross-fitting

## 1 Introduction

A central challenge in applied causal inference is determining whether an estimate obtained from nonrandomized data can recover a causal effect that would be estimated under random assignment. Within-study comparisons (WSCs) address this challenge by comparing randomized and nonrandomized estimates in a shared study setting, reducing the extent to which differences across studies, populations, interventions, or outcome measurements explain discrepancies between estimates (Shadish et al., 2008; Keller et al., 2025). The design is valuable because it provides an empirical

benchmark, while still requiring care in interpreting the benchmark and the assumptions needed for observational identification.

The four-arm mathematics–vocabulary WSC described by Keller et al. (2025) combines randomized assignment and self-selection indicators to construct randomized and quasi-experimental comparisons. The public WSCdata release makes the data available for further methodological work. The present analysis uses this resource to examine a narrow question: how do a protocol-aligned randomized contrast and an observational estimate of the corresponding target compare in the available sample, and what can a separate full-sample ATE comparison add to that assessment?

Three concepts must be distinguished. First, *parameter equivalence* is a design-level statement that two estimands refer to the same causal parameter under specified assumptions. Second, *observational identification* concerns whether the observational functional identifies that parameter from measured data; it generally requires assumptions such as conditional exchangeability, positivity, and consistency. Third, *estimation uncertainty* concerns how precisely the parameter is estimated in the finite sample. Parameter equivalence does not establish observational identification, and similar point estimates do not establish equivalence or precise agreement.

The frozen protocol defines a primary comparison between the randomized mathematics effect among participants who selected mathematics and a cross-fitted AIPW ATT in the concordant QED subset. The protocol's design argument, based on Section 4 of Keller et al. (2025), states that these targets are equivalent under random assignment, positivity, and the specified group definitions. This equivalence does not itself establish the conditional exchangeability or treatment-version consistency needed to identify the observational target. The protocol also forbids interpreting interval overlap as equivalence and specifies no equivalence margin.

A secondary full-sample ATE comparison was subsequently authorized by an explicit owner decision. Its estimator choices were made after the primary results were known but before the secondary estimates were examined. It is therefore a secondary analysis and the overall study is not fully preregistered. In addition, the design probability of one-half required by the cited ATE parameter-equivalence argument was not independently verified in the available data documentation. Accordingly, the secondary pair is reported descriptively and is not treated as two estimates of a proven common parameter.

This paper makes a deliberately bounded contribution. It reports the two authorized comparisons, their uncertainty, and diagnostics for overlap and measured-covariate balance. It does not attempt to estimate a general "degree of estimand alignment," establish that alignment caused any difference between the observed contrasts, or conclude that observational estimates have recovered randomized effects. Only the mathematics post-test outcome is analyzed here; vocabulary post-test, ATU, and additional estimator pairings are outside the authorized scope.

# 2 Study Design and Data

## 2.1 Data source and outcome

The analysis uses the WSCdata 0.1.2 release associated with the four-arm WSC described by Keller et al. (2025). The study includes 2,200 adult participants recruited online and examines mathematics training relative to a vocabulary session, with baseline measures and post-test outcomes. The primary outcome in the present analysis is the mathematics post-test score, `mathPost`. The vocabulary post-test is not combined with the mathematics outcome and is not analyzed in this

paper.

The source data encode the mathematics-group indicator `mathGrp` and the mathematics-selection indicator `mathSel`. Let $D_i$ denote the analysis treatment indicator for mathematics, coded from `mathGrp`, and let $S_i$ denote `mathSel`. The four cells used in the frozen protocol are shown in Table 1. No additional sample exclusions were introduced for the reported comparisons.

**Table 1:** Four cells defined by the mathematics group and selection indicators

| Cell | `mathSel` | `mathGrp` | $n$ |
|---|---|---|---|
| G1 | 1 | 1 | 288 |
| G2 | 1 | 0 | 319 |
| G3 | 0 | 1 | 776 |
| G4 | 0 | 0 | 817 |
| Total | | | 2,200 |

*Note.* The observed mathematics-group count is 1,064 (G1 + G3), and the comparison-group count is 1,136 (G2 + G4). These realized counts are used as observed; the randomized estimator is not reweighted to impose a 50/50 allocation.

### 2.2 Analysis populations

The primary randomized benchmark uses G1 ∪ G2, the 607 participants with `mathSel` = 1 (288 in G1 and 319 in G2). It estimates the randomized mathematics effect among mathematics-preferring participants. The primary observational comparison uses G1 ∪ G4, the 1,105 concordant observations for which `mathGrp` = `mathSel` (288 in G1 and 817 in G4). In this subset, the treated observations are G1 and the comparison observations are G4. Because `mathSel` equals the treatment indicator within this subset, it is not included as a separate adjustment covariate.

The secondary randomized ATE uses all 2,200 observations. The secondary observational ATE uses G1 ∪ G4, as in the primary observational analysis. The randomized and observational samples therefore differ in the secondary comparison, and the primary and secondary comparisons target different estimands.

## 3 Estimands and Identification Conditions

### 3.1 Primary, protocol-aligned comparison

Let $Y_i(1)$ and $Y_i(0)$ denote potential mathematics post-test scores under mathematics and vocabulary treatment, respectively. Let $S_i = 1$ indicate the mathematics-selection group. The primary randomized estimand is

$$\tau_{\text{RCT},S=1} = \mathbb{E}[Y(1) - Y(0) \mid S = 1]\,, \tag{1}$$

estimated in G1 ∪ G2 by the difference in observed means across the randomized mathematics-group indicator.

The aligned observational target is the ATT in the concordant subset:

$$\tau_{\text{QED,ATT}} = \mathbb{E}[Y(1) - Y(0) \mid D = 1,\ i \in \text{G1} \cup \text{G4}]\,. \tag{2}$$

The frozen protocol's design argument states that Equations (1) and (2) refer to the same causal parameter under random assignment, positivity for selection and assignment, and the group definitions in the WSC design (Keller et al., 2025). The ATT equivalence argument does not require a one-half treatment-assignment probability.

This design-level equivalence must not be confused with observational identification. For the QED functional to identify the ATT, the analysis additionally requires, within G1 ∪ G4, conditional exchangeability of treatment and potential outcomes given the prespecified covariates, positivity of treatment conditional on those covariates, and consistency. Consistency includes the assumption that randomized assignment and self-selection do not represent substantively different versions of treatment. Conditional exchangeability, especially with respect to unmeasured factors, cannot be established from covariate balance alone.

### 3.2 Secondary full-sample ATE comparison

The secondary randomized target is the full-sample ATE,

$$\tau_{\text{RCT,full}} = \mathbb{E}[Y(1) - Y(0)]\,, \tag{3}$$

estimated as the unadjusted difference in mean `mathPost` between the observed mathematics and comparison groups across all 2,200 observations.

The secondary observational target is the ATE in the concordant subset,

$$\tau_{\text{QED,ATE}} = \mathbb{E}[Y(1) - Y(0) \mid i \in \text{G1} \cup \text{G4}]\,. \tag{4}$$

The design-level proof cited in the frozen protocol requires $\mathbb{P}(D = 1) = \mathbb{P}(D = 0) = 1/2$ for the full-sample RCT ATE and concordant-subset QED ATE to be equivalent, in addition to the relevant random-assignment and observational-identification conditions. The available WSCdata documentation confirms the randomized-group coding but does not independently document the one-half assignment probability. The realized counts of 1,064 and 1,136 do not establish that design condition. Consequently, this paper does not assert that Equations (3) and (4) are the same parameter.

## 4 Statistical Methods

### 4.1 Randomized difference-in-means estimators

For each randomized benchmark, the point estimate is the unadjusted difference in mean `mathPost` scores between the mathematics group and the comparison group in the specified analysis population:

$$\widehat{\tau}_{\text{RCT}} = \overline{Y}_{D=1} - \overline{Y}_{D=0}. \tag{5}$$

The Neyman standard error is

$$\widehat{\text{SE}}(\widehat{\tau}_{\text{RCT}}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}, \tag{6}$$

where $s_d^2$ is the sample variance in group $d$, calculated with denominator $n_d - 1$. Individual randomized-estimate intervals are 95% normal intervals using the standard normal critical value. The full-sample estimator uses the observed treatment-group counts and does not reweight the two observed means to 50/50.

### 4.2 Cross-fitted AIPW estimators

The observational estimates use five-fold cross-fitted AIPW with main-effects logistic regression for the propensity score and main-effects linear outcome regressions. The same prespecified set of 25 baseline covariates is used: `female`, `white`, `black`, `asian`, `hisp`, `married`, `logAge`, `income`, `collegeS`, `collegeM`, `collegeD`, `calc`, `logBooks`, `mathLike`, `big5O`, `big5C`, `big5E`, `big5A`, `big5N`, `AMAS`, `logBDI`, `MCS`, `GSES`, `vocabPre`, and `mathPre`. The treatment indicator, post-treatment outcomes, row identifier, and raw versions of prespecified log-transformed covariates are not included in the adjustment set. In the concordant subset, `mathSel` is excluded because it is identical to treatment.

For the ATT, let $e(X) = \mathbb{P}(D = 1 \mid X)$, $\widehat{e}(X)$ be its out-of-fold estimate, $\widehat{\mu}_0(X)$ be the estimated control outcome regression, and $\widehat{p} = n^{-1}\sum_i D_i$. The cross-fitted ATT score used in the implementation is

$$\widehat{\tau}_{\mathrm{ATT}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{D_i}{\widehat{p}}\{Y_i - \widehat{\mu}_0(X_i)\} - \frac{(1-D_i)\widehat{e}(X_i)}{\widehat{p}\{1-\widehat{e}(X_i)\}}\{Y_i - \widehat{\mu}_0(X_i)\}\right]. \tag{7}$$

The ATT score targets the treated observations in the concordant subset. Its implementation uses the control outcome regression in the ATT score.

For the ATE, the cross-fitted score is

$$\widehat{\tau}_{\mathrm{ATE}} = \frac{1}{n}\sum_{i=1}^{n}\left[\widehat{\mu}_1(X_i) - \widehat{\mu}_0(X_i) + \frac{D_i\{Y_i - \widehat{\mu}_1(X_i)\}}{\widehat{e}(X_i)} - \frac{(1-D_i)\{Y_i - \widehat{\mu}_0(X_i)\}}{1-\widehat{e}(X_i)}\right], \tag{8}$$

where $\widehat{\mu}_1(X)$ and $\widehat{\mu}_0(X)$ are the fitted mathematics and comparison outcome regressions. These are standard augmented inverse-probability-weighted forms; double robustness is conditional on the identifying assumptions and does not remove unmeasured confounding (Bang and Robins, 2005).

Five-fold cross-fitting uses shuffle seed 20261026. In bootstrap replications, copies of the same original participant are kept in the same fold so that a participant's copies do not appear in both the training and test folds. Propensity predictions are numerically bounded to $[10^{-6}, 1-10^{-6}]$ before entering the scores. No observation was removed because of overlap and no scientific trimming was applied. The interval $[0.1, 0.9]$ is used only as a descriptive overlap window.

The individual QED standard errors are influence-function standard errors, calculated as the sample standard deviation of the fitted score (using $ddof = 1$) divided by the square root of the analysis sample size. Individual-estimate 95% intervals are normal intervals. They are distinct from the bootstrap intervals for paired contrasts.

### 4.3 Paired contrasts and bootstrap uncertainty

For the primary analysis, the contrast is

$$\Delta_{\mathrm{ATT}} = \tau_{\mathrm{QED,ATT}} - \tau_{\mathrm{RCT},S=1}. \tag{9}$$

The joint participant bootstrap uses 2,000 replications and seed 20261026. Participants are resampled with replacement within G1, G2, and G4, preserving each observed cell size. The same resampled G1 observations enter both estimators. G3 is not part of either primary estimator. Each replication recomputes the randomized mean difference and refits the observational nuisance models and cross-fitted ATT score. The contrast is calculated within each replication. The reported uncertainty

includes the bootstrap standard error and the 2.5th and 97.5th percentiles of the bootstrap contrast distribution.

For the secondary analysis, the contrast is

$$\Delta_{\text{ATE}} = \tau_{\text{QED,ATE}} - \tau_{\text{RCT,full}}. \tag{10}$$

The secondary joint bootstrap resamples with replacement within all four cells, preserving the observed cell sizes. The same G1 and G4 draws are shared across the two estimators; G3 enters only the full-sample randomized estimator. Each replication refits the propensity and outcome models and recomputes the contrast.

Both procedures completed all 2,000 requested replications, with no failures and no replacement of failed replications. The resampling procedures condition on the observed cell counts; they are not randomization-inference procedures. Successful completion does not establish nominal interval coverage or causal identification. The primary bootstrap was specified in the frozen confirmatory protocol. The secondary estimator and its zero-failure reporting rule were chosen in a separate owner decision after the primary results were known but before the secondary estimates were examined.

### 4.4 Overlap, effective sample size, and balance

For the ATT, control weights are proportional to $\widehat{e}(X)/(1-\widehat{e}(X))$. For the ATE, treatment and control weights are $1/\widehat{e}(X)$ and $1/\{1-\widehat{e}(X)\}$, respectively. Effective sample size is computed as

$$\widehat{\text{ESS}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{11}$$

Overlap is summarized by the propensity-score range and the number of observations outside $[0.1, 0.9]$. This window is descriptive and does not define an exclusion rule.

Covariate balance is summarized using standardized mean differences, with the denominator fixed to the unweighted pooled standard deviation in the G1/G4 subset. The values 0.1 and 0.2 are descriptive reference points only; they are not pass/fail criteria and were not used to modify the analysis specification.

## 5 Results

### 5.1 Primary comparison: randomized effect among mathematics-preferring participants versus QED ATT

The randomized estimate in G1 ∪ G2 was 0.794681731104145 (Neyman SE 0.2777657065981394; normal 95% CI [0.25027095003147215, 1.3390925121768178]). The cross-fitted QED ATT in G1 ∪ G4 was 0.7942342079185714 (influence-function SE 0.22379148303184127; normal 95% CI [0.35561096112935586, 1.2328574547077868]). The point estimates are close, but the estimates use different analysis populations and different estimators; they share the G1 treated group, while their comparison groups differ.

The original-sample contrast, QED ATT minus randomized estimate, was −0.0004475231855736172 5. Its joint-bootstrap standard error was 0.4216686503551098 and its 95% percentile interval was [−0.9491600721638183, 0.6638487257328824]. All 2,000 bootstrap replications succeeded. The

interval includes zero and also values representing appreciable differences in either direction. The small point-estimate difference therefore does not establish equivalence, negligible bias, or precise agreement.

**Table 2:** Primary comparison for mathematics post-test scores

| Estimate | $n$ | Point estimate | SE | 95% interval |
|---|---|---|---|---|
| Randomized effect among math-preferring participants (G1 ∪ G2) | 607 | 0.794682 | 0.277766 | [0.250271, 1.339092] |
| QED concordant ATT (G1 ∪ G4) | 1,105 | 0.794234 | 0.223791 | [0.355611, 1.232857] |
| Paired contrast: QED ATT − RCT | – | −0.000448 | 0.421669 | [−0.949160, 0.663849] |

*Note.* The intervals for the two individual estimates are normal intervals. The interval for the paired contrast is the 2,000-replicate joint-bootstrap percentile interval. Full-precision values are listed in Appendix A. The randomized estimate uses G1 ∪ G2; the QED estimate uses G1 ∪ G4.

## 5.2 Overlap and measured-covariate diagnostics for the primary QED estimate

The out-of-fold propensity scores in G1 ∪ G4 ranged from 0.00965643798321191 to 0.921102740162679. There were 396 of 1,105 observations (35.84%) outside the descriptive $[0.1, 0.9]$ window. No observations were removed and no propensity predictions were moved by the numerical floor.

The ATT control-weight effective sample size was 145.91819610069288 out of 817 controls (17.86%). The maximum control weight was 7.7029890726271155. Before weighting, the largest absolute standardized mean difference was 0.994464628090502 for `mathLike`; 18 of 25 covariates had absolute standardized differences above 0.1 and eight were above 0.2. After weighting, no covariate exceeded 0.2, but three remained above 0.1: `vocabPre` (−0.14297294382753992), `asian` (−0.13030869797577316), and `income` (0.11946662392917676). These diagnostics are descriptive. Improved measured-covariate balance does not establish the absence of unmeasured confounding.

## 5.3 Secondary comparison: full-sample randomized ATE versus concordant-subset QED ATE

The full-sample randomized ATE estimate was 0.7603449645239859 (Neyman SE 0.1510881033486582; normal 95% CI [0.4642177234681503, 1.0564722055798215]) using all 2,200 observations, including 1,064 assigned to mathematics and 1,136 assigned to the comparison group. The QED concordant-subset ATE estimate was 1.0753974159386432 (influence-function SE 0.23320145971183331; normal 95% CI [0.6183309537612816, 1.532463878116005]) using 1,105 observations (288 treated and 817 controls).

The original-sample difference, QED ATE minus full-sample RCT ATE, was 0.3150524514146573. The bootstrap standard error was 0.45778170858020434 and the 95% percentile interval was [−0.6249039850074877, 1.1444295170363872]. All 2,000 requested replications succeeded. The interval includes zero and positive differences of appreciable size; it neither establishes equality nor establishes a difference between a proven common parameter.

**Table 3:** Secondary full-sample ATE comparison for mathematics post-test scores

| Estimate | $n$ | Point estimate | SE | 95% interval |
|---|---|---|---|---|
| Full-sample randomized ATE | 2,200 | 0.760345 | 0.151088 | [0.464218, 1.056472] |
| QED concordant-subset ATE | 1,105 | 1.075397 | 0.233201 | [0.618331, 1.532464] |
| Paired contrast: QED ATE − RCT ATE | – | 0.315052 | 0.457782 | [−0.624904, 1.144430] |

*Note.* The individual-estimate intervals are normal intervals; the paired contrast interval is a joint-bootstrap percentile interval. The secondary estimator choices were made after the primary results were known and before the secondary estimates were examined. This analysis does not replace the primary analysis and does not make the study fully preregistered.

### 5.4 Overlap and measured-covariate diagnostics for the secondary QED ATE

The secondary ATE analysis used the same 1,105-person concordant subset and the same descriptive propensity window. Again, 396 observations (35.84%) fell outside $[0.1, 0.9]$, and no observations were trimmed. The ATE-weight effective sample size was 89.1696953554412 out of 288 treated observations and 603.2355199300506 out of 817 controls. The maximum treatment weight was 53.5720736802485, indicating substantial concentration in the treated weights.

Before weighting, the largest absolute standardized mean difference was 0.994464628090502 for `mathLike`. After weighting, the largest absolute difference was 0.22220296200610962 for `big5A`; three covariates had absolute weighted differences above 0.1 (`calc`, `mathLike`, and `big5A`), and one exceeded 0.2. These findings reinforce that the ATE weighting distribution is concentrated and that measured-covariate balance is not perfect. Neither the weighting diagnostics nor the AIPW construction resolves unmeasured confounding.

**Table 4:** Overlap and effective sample size in the concordant QED subset

| Diagnostic | Primary ATT | Secondary ATE |
|---|---|---|
| Concordant-subset sample size | 1,105 | 1,105 |
| Observations outside $[0.1, 0.9]$ | 396 (35.84%) | 396 (35.84%) |
| Treatment ESS | – | 89.17 / 288 |
| Control ESS | 145.92 / 817 | 603.24 / 817 |
| Maximum treatment weight | – | 53.57 |
| Maximum control weight | 7.70 | 8.70 |
| Largest absolute weighted SMD | No value above 0.2 | 0.2222 (`big5A`) |

*Note.* ESS is computed from the estimator-specific weights. The propensity window is descriptive only. SMD thresholds are not pass/fail criteria. Full-precision diagnostics are available in the project result files.

## 6 Discussion

### 6.1 What the primary comparison shows

The primary point estimates are nearly identical in the observed sample: the randomized estimate is 0.794681731104145 and the QED ATT estimate is 0.7942342079185714. This is a descriptive feature of the point estimates, not a demonstration that the observational procedure recovered the randomized causal effect. The joint-bootstrap interval for the contrast is wide, extending

from −0.9491600721638183 to 0.6638487257328824. No equivalence margin was specified, and no equivalence test was conducted. The interval's inclusion of zero is not evidence of equivalence.

The distinction between parameter equivalence and observational identification is central to interpretation. The design argument in Keller et al. (2025) supports a claim that the two primary estimands refer to the same causal parameter under its assumptions. It does not show that the QED AIPW functional is identified in this sample. Identification additionally depends on conditional exchangeability, positivity, and consistency in the concordant subset. The analysis cannot verify the absence of unmeasured confounding, and the residual measured-covariate imbalance and weight concentration counsel against interpreting the close point estimates as a definitive validation result.

### 6.2 What the secondary ATE comparison adds

The secondary comparison produces a larger point-estimate difference: 1.0753974159386432 for the concordant-subset QED ATE versus 0.7603449645239859 for the full-sample randomized ATE. However, its bootstrap interval, [−0.6249039850074877, 1.1444295170363872], is also wide and includes zero. It does not establish that the estimates differ, and it does not establish that they are equivalent.

More importantly, the secondary pair differs from the primary pair in target population, estimand, and observational estimator. The primary randomized benchmark is restricted to the mathematics-selection group and is paired with an ATT estimator. The secondary randomized benchmark uses the full sample and is paired with an ATE estimator in the concordant subset. Therefore, the difference between the two observed contrasts cannot be attributed solely to estimand alignment, nor can either contrast be interpreted as a pure measure of confounding bias.

The secondary ATE comparison has an additional design limitation. The parameter-equivalence argument cited in the frozen protocol requires a one-half treatment-assignment probability, which the available data documentation did not independently verify. The observed allocation of 1,064 versus 1,136 is a realized finite-sample count, not independent evidence of the design probability. For that reason, the paper does not claim that the secondary randomized and observational estimators target the same parameter.

### 6.3 Overlap, weighting, and double robustness

The overlap diagnostics show that 35.84% of observations in the concordant subset lie outside the descriptive propensity window. For the primary ATT, the control-weight ESS is approximately 17.9% of the 817 available controls. For the secondary ATE, the treatment-weight ESS is approximately 31.0% of the 288 treated observations, and the maximum treatment weight is above 53. These diagnostics indicate that the weighted estimates rely disproportionately on subsets of observations and may be imprecise.

The measured-covariate balance improves after weighting on many variables but is not uniformly close to zero. The ATE-weighted `big5A` standardized difference is approximately 0.222, while the primary ATT retains absolute standardized differences above 0.1 for `vocabPre`, `asian`, and `income`. The descriptive thresholds are not formal acceptance criteria. Even perfect balance on measured covariates would not demonstrate conditional exchangeability with respect to unmeasured covariates.

AIPW combines an outcome model and a propensity model, and its double-robust property can protect consistency if the relevant identifying assumptions hold and at least one nuisance model is

correctly specified (Bang and Robins, 2005). It does not remove unmeasured confounding, solve treatment-version inconsistency, or make a poorly supported overlap region informative. Cross-fitting reduces dependence between nuisance-model fitting and score evaluation, but it likewise does not replace the causal assumptions (Chernozhukov et al., 2018).

### 6.4 Preregistration and scope

The primary contrast was governed by the frozen protocol and the associated owner approval. The secondary ATE estimators were selected in a separate owner decision after the primary result was known but before the secondary estimates were examined. This chronology is disclosed rather than characterized as full preregistration. The secondary analysis does not replace the primary analysis or retrospectively select the primary specification.

The analysis is limited to `mathPost` and the two completed pairs. The vocabulary outcome, ATU, L3, alternative estimators, and additional cross-pairings were not run and are not part of the present evidence. The results therefore do not answer the broader question of how much estimand alignment improves agreement between observational and randomized estimates. Addressing that question would require design evidence for the relevant parameter-equivalence conditions and a comparison criterion fixed independently of the observed contrasts.

## 7 Limitations

Several limitations constrain interpretation.

1. **Observational identification is not established.** Conditional exchangeability within the concordant subset is not testable from the observed data alone. Unmeasured confounding remains possible, and the analysis does not establish consistency across randomized and self-selected treatment versions.
2. **Overlap and effective sample size are limited.** A substantial share of the concordant subset lies outside the descriptive propensity window, and the ATT control weights and ATE treatment weights are concentrated. No scientific trimming was applied, so the reported estimates retain the protocol-defined target rather than changing it to address poor overlap.
3. **The ATE parameter-equivalence condition is unverified.** The available documentation does not independently establish a one-half assignment probability. Consequently, the secondary ATE estimates are not asserted to target the same parameter.
4. **The comparisons are not interchangeable.** The primary and secondary pairs differ in population, estimand, and estimator. The change in their numerical contrasts cannot be interpreted as the effect of estimand alignment or as a direct measure of confounding.
5. **The study is not fully preregistered.** The secondary estimator choices were made after the primary results were known, although before the secondary estimates were examined. The secondary analysis is explicitly labeled as such.
6. **The scope is narrow.** Only mathematics post-test scores and two authorized estimator pairs are reported. No claim is made about the vocabulary outcome, ATU, L3, other estimators, or other WSC settings.

7. **A code-provenance limitation remains.** The technical audit found that the current `aipw.py` file has a different SHA-256 hash from the execution-time version recorded for the primary analysis. The difference followed later additions for ATE estimation and diagnostics. The exact historical bytes were not retained, so a byte-for-byte comparison was not possible. The audit reviewed the current ATT score against the frozen specification and found no estimator discrepancy requiring a rerun, but exact reconstruction of the primary implementation remains unverified.

# 8 Conclusion

In the primary, design-aligned comparison, the randomized mathematics effect among mathematics-preferring participants and the concordant-subset QED ATT had nearly identical point estimates. However, the joint-bootstrap interval was wide, so the results do not establish equivalence, negligible confounding, or recovery of the randomized causal effect. In the secondary full-sample ATE comparison, the point estimates were farther apart, but the uncertainty interval also included zero and the design condition needed to assert parameter equivalence was not independently verified.

Taken together, the two comparisons illustrate why point-estimate proximity, parameter equivalence, observational identification, and precision must be treated as separate questions. They do not identify the degree to which estimand alignment causes observational estimates to approach randomized benchmarks. The evidence supports reporting these bounded comparisons with explicit uncertainty and assumptions, while leaving the broader alignment question unresolved.

# Data, Code, and Reproducibility Statement

The source data are available through the WSCdata R package, version 0.1.2 (Keller et al., 2025). The analysis used the frozen protocol and stored machine-readable result files. The source-data SHA-256 recorded by the project is `b47be7d8f2c0d54af9cc71f3602183e61307a964e701a2cdcf3944bcef85286b`; the frozen-protocol SHA-256 is `583ff75e5e848e4c1831e6dbc6084041afec44c5a18a4806d31db6f3d5564e59`. No real-data estimator or bootstrap was rerun for manuscript preparation. The numerical values in this manuscript were transcribed from the stored analysis reports and their audit.

A public code archive and a stable release of the machine-readable results should be supplied before submission. The exact historical bytes of the primary `aipw.py` implementation were not preserved; this limitation is described above and should remain visible unless resolved with an exact hash match. Ethics approval and consent details should be reported from the original study documentation before submission; they were not independently established from the analysis files used for this draft.

# A Full-Precision Numerical Record

The main tables round estimates and standard errors for readability. The values below preserve the full-precision numerical literals recorded in the project's analysis reports.

## A.1 Primary comparison

| Quantity | Full-precision value |
|---|---|
| Randomized estimate, `RCT_effect_prefer_math` | 0.794681731104145 |
| Neyman standard error | 0.2777657065981394 |
| Normal 95% CI, lower | 0.25027095003147215 |
| Normal 95% CI, upper | 1.3390925121768178 |
| Randomized sample size | 607 (288 treated; 319 controls) |
| QED estimate, `QED_concordant_ATT` | 0.7942342079185714 |
| Influence-function standard error | 0.22379148303184127 |
| Normal 95% CI, lower | 0.35561096112935586 |
| Normal 95% CI, upper | 1.2328574547077868 |
| QED sample size | 1105 (288 treated; 817 controls) |
| Contrast (QED ATT − RCT) | -0.00044752318557361725 |
| Joint-bootstrap standard error | 0.4216686503551098 |
| Bootstrap percentile 95% CI, lower | -0.9491600721638183 |
| Bootstrap percentile 95% CI, upper | 0.6638487257328824 |
| Bootstrap replications / successes / failures | 2000 / 2000 / 0 |
| Bootstrap covariance of estimates | 0.0064289260006763025 |
| Propensity-score minimum / maximum | 0.00965643798321191 / 0.921102740162679 |
| Observations outside $[0.1, 0.9]$ | 396 / 1105 = 0.3583710407239819 |
| ATT control-weight ESS | 145.91819610069288 / 817 = 0.17860244320770244 |
| Maximum ATT control weight | 7.7029890726271155 |
| Maximum absolute unweighted SMD | 0.994464628090502 (`mathLike`) |
| Weighted SMDs with absolute value above 0.1 | vocabPre: -0.14297294382753992; asian: -0.13030869797577316; income: 0.11946662392917676 |

## A.2 Secondary full-sample ATE comparison

| Quantity | Full-precision value |
|---|---|
| Full-sample randomized estimate, `RCT_ATE_full` | 0.7603449645239859 |
| Neyman standard error | 0.1510881033486582 |
| Normal 95% CI, lower | 0.4642177234681503 |
| Normal 95% CI, upper | 1.0564722055798215 |
| Randomized sample size | 2200 (1064 treated; 1136 controls) |
| QED estimate, `QED_concordant_ATE` | 1.0753974159386432 |
| Influence-function standard error | 0.23320145971183331 |

| Quantity | Full-precision value |
|---|---|
| Normal 95% CI, lower | 0.6183309537612816 |
| Normal 95% CI, upper | 1.532463878116005 |
| QED sample size | 1105 (288 treated; 817 controls) |
| Contrast (QED ATE − full-sample RCT ATE) | 0.3150524514146573 |
| Joint-bootstrap standard error | 0.45778170858020434 |
| Bootstrap percentile 95% CI, lower | -0.6249039850074877 |
| Bootstrap percentile 95% CI, upper | 1.1444295170363872 |
| Bootstrap replications / successes / failures | 2000 / 2000 / 0 |
| Bootstrap covariance of estimates | 0.00785036882408469 |
| Treatment-weight ESS | 89.1696953554412 / 288 = 0.3096169977619486 |
| Control-weight ESS | 603.2355199300506 / 817 = 0.7383543695594255 |
| Maximum treatment weight | 53.5720736802485 |
| Maximum control weight | 8.702989072627116 |
| Maximum absolute unweighted SMD | 0.994464628090502 (mathLike) |
| Maximum absolute weighted SMD | 0.22220296200610962 (big5A) |
| Weighted SMDs with absolute value above 0.1 | calc: -0.1153628017786516; mathLike: -0.12406260273575813; big5A: 0.22220296200610962 |

# B Protocol and analysis chronology

The primary contrast was the only confirmatory comparison under the frozen protocol. The secondary ATE comparison was separately authorized on October 27, 2026. Its estimator choices were not uniquely specified by the frozen protocol, and the analysis is not described as fully preregistered. Both analyses completed 2,000 requested bootstrap replications with zero failures. The technical audit on October 28, 2026, found no estimator implementation error requiring a rerun, but rated the overall audit as a partial pass because the exact historical bytes of the primary `aipw.py` implementation were not retained. No L3, `vocabPost`, ATU, or additional estimator pairing was run for this manuscript.

# References


Bang, H., and J. M. Robins. (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4), 962–973. https://doi.org/10.1111/j.1541-0420.2005.00377.x.

Chernozhukov, V., D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1), C1–C68. https://doi.org/10.1111/ectj.12097.

Keller, B., V. C. Wong, S. Park, J. Zhang, P. Sheehan, and P. M. Steiner. (2025). A new four-arm within-study comparison: Design, implementation, and data. *Observational Studies*, 11(2), 153–188. https://doi.org/10.1353/obs.2025.a963647.

Keller, B., S. Park, and J. Zhang. (2025). *WSCdata: A new four-arm within-study comparison data on math and vocabulary training* (R package version 0.1.2). https://CRAN.R-project.org/package=WSCdata.

Rosenbaum, P. R., and D. B. Rubin. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1), 41–55. https://doi.org/10.1093/biomet/70.1.41.

Shadish, W. R., M. H. Clark, and P. M. Steiner. (2008). Can nonrandomized experiments yield accurate answers? A randomized experiment comparing random and nonrandom assignments to treatments. *Journal of the American Statistical Association*, 103(484), 1334–1356. https://doi.org/10.1198/016214508000000733.